A three-step least-squares iterative method for tilt phase-shift interferometry

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Abstract: An iterative method based on least-squares fittings is proposed to retrieve wavefront phase from tilt phase-shift interferograms. In each iteration cycle, proposed method calculates wavefront phase and tilt phase shifts in *x*- and *y*-directions in three individual least-squares fitting steps. In tilt phase shifts extracting steps, phase shifts of interferograms columns or rows are calculated with least-squares method, and then tilt phase shifts of interferograms in *x*- or *y*-direction are determined by linear regressions. At least three interferograms of three by three pixels are required with proposed method. The performance of proposed method is demonstrated by simulations and experiments. Tilt gradients and translational phase shifts could be extracted with high accuracy and large wavefront tilts could be well handled with proposed method. The method could be applied to temporal phase-shift interferometers with uncalibrated transducers or that in vibrating environment.

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OCIS codes: (120.3180) Interferometry; (120.5050) Phase measurement; (050.5080) Phase shift.

References and links

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1. Introduction

Temporal phase-shift (TPS) interferometer is regarded as a powerful instrument for wavefront phase measurement. For its high accuracy, high spatial resolution and good noise immunity, TPS interferometer has achieved widespread use in a variety of interferometric applications. In implementation of TPS interferometry, phase-shifting device, i.e. mechanical transducer or wavelength-tuned laser, generates phase shifts and at least three interferograms should be collected sequentially in time domain. Phase shifts are expected to be uniform at all pixels in an interferogram, usually named piston phase shifting. However, during phase shifting process, environmental vibrations or imperfect phase-shifting device will change the relative orientation between reference and test wavefronts. Expected piston phase shifts are disturbed and phase shifts at pixels in an interferogram are not uniform any longer. The non-uniform phase shifts, named tilt phase shifts, will cause tilt-shift errors in TPS interferometry. In tilt phase-shift interferometry, translational phase shifts and tilts gradients in x- and y-directions are required to extract to compensate tilt-shift errors.

Many methods have been developed to cope with interferograms with tilt phase shifts. Among these methods, some of them are based on spatial analysis of interferograms [1–4]. They may suffer from low flexibility [1–3] or retrace errors [4–6]. Some method [7,8] assumes that the intensity variations are ergodic in temporal domain. But large number (about 100 frames) interferograms are required to reduce uncertainty of parameters estimations. Besides, an important method is least-squares iterative method. This method could obtain solutions of unknowns in interferometry mathematically, including wavefront phase and phase shifts. Okada *et al* [9] first proposed least-squares iterative algorithm. Then Han and Kim [10] proposed another iterative algorithm. In 2004, Wang and Han [11] made improvement and developed Advanced Iterative Algorithm. With Wang's algorithm, the iteration converges faster and more accurately, and requires lower accuracy of initial phase shift estimations. And only three interferograms are required at least. These above iterative methods [9–11] could only deal with random phase-shift interferograms but tilt phase-shift interferograms. Nonlinearity of interferogram intensity equation increases the inconvenience of determining tilt phase shifts with least-squares iterative methods. To compensate tilt phase shifts, Chen *et al* [12] expanded interferogram intensity in first-order Taylor series expansion to replace the nonlinear equation in iterative algorithm. Due to the approximations of first-order Taylor series expansion, Chen's method has obvious errors when large wavefront tilts are present in interferograms. Xu *et al* [13] divided interferograms to small regions and make assumption that tilt phase shifts are neglect in each region. Then local phase shifts of regions are extracted with Wang's algorithm, and tilt phase-shift planes are fitted. Then tilt phase shifts are then compensated over all pixels of interferograms. Xu's method is more flexible than Chen's method. However, due to the assumption an inherent error is in presence. Besides, the region numbers should be assigned manually in different cases. Chen *et al* [14] solved nonlinearity of tilt phase shift in iterations by performing nonlinear least-squares fittings. Chen's method could overcome tilt-shift errors well, but an approximation is made that background and modulation are uniform when tilt phase shifts are estimated.

In this paper, we present a three-step iterative method immune to tilt phase shifts. Each iterative cycle is composed of three steps, in which wavefront phase and tilt phase shifts in x- and y-directions are determined with linear least-squares method. In the second step, phase shifts of interferogram columns are calculated and then tilt phase shifts in x-direction are determined by linear regressions. Likewise, in the third step, tilt phase shifts in y-direction could be determined after phase shifts of rows are calculated. The extraction of column and row phase shifts is based on Wang's algorithm. For no additional assumption is made in calculating tilt phase shifts, our method estimates phase shifts more accurately. Therefore, large tilt phase shifts could be extracted with our method. Principle of our method is exhibited, and simulations and experiments demonstrate its performance.

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2. Algorithm descriptions

In TPS interferometry, the intensity of the mth interferograms could be expressed as

$$\begin{aligned} I_m{}^{t}(x,y) &= A(x,y) + B(x,y)\cos[\varphi(x,y) + k_{xm}x + k_{ym}y + \delta_m] \\ &= A(x,y) + B(x,y)\cos[\varphi(x,y) + \Delta_{xm}(x) + \Delta_{ym}(y)], \end{aligned} \tag{1a}$$

where $A(x,y)$, $B(x,y)$ and $\varphi(x,y)$ are the background, amplitude of modulation and wavefront phase at pixel (x,y) respectively, and k_{xm}, k_{ym} and δ_m are x- and y-directional tilt gradients and translational phase shift of phase-shift plane in the mth interferogram. The superscript t denotes theoretical value. The x- and y-directional components of tilt phase shifts are defined as $\Delta_{xm}(x)$ and $\Delta_{ym}(y)$ with relationships:

$$\begin{cases} \Delta_{xm}(x) = k_{xm}x + \delta_{xm} \\ \Delta_{ym}(y) = k_{ym}y + \delta_{ym}. \\ \delta_m = \delta_{xm} + \delta_{ym} \end{cases} \tag{1b}$$

The algorithm is composed of three steps and is described as following.

Step 1. Determine wavefront phase with x- and y-directional tilt phase shifts known

In this step, it's assumed that tilt gradients k_{xm} and k_{ym} and translational phase shifts δ_{xm} and δ_{ym} are known. It means that phase shift of every pixel (x,y) is known. The theoretical intensity of interferogram is

$$I_m{}^{t}(x,y) = a(x,y) + b(x,y)\cos[\Delta_m(x,y)] + c(x,y)\sin[\Delta_m(x,y)], \tag{2}$$

where $a(x,y) = A(x,y)$, $b(x,y) = B(x,y)\cos[\varphi(x,y)]$, $c(x,y) = -B(x,y)\sin[\varphi(x,y)]$ are unknowns, and $\Delta_m(x,y) = \Delta_{xm}(x) + \Delta_{ym}(y)$ is known. The least-squares errors for three unknowns, $a(x,y)$, $b(x,y)$ and $c(x,y)$, can be expressed as

$$S(x,y) = \sum_{m=1}^{M}[I_m{}^{t}(x,y) - I_m(x,y)]^2, \tag{3}$$

where $I_m(x,y)$ is experimental intensity of the mth interferogram and M is total number of interferograms. With $\Delta_m(x,y)$ known, the least-squares criteria are expressed as

$$\frac{\partial S(x,y)}{\partial a(x,y)} = 0, \frac{\partial S(x,y)}{\partial b(x,y)} = 0, \frac{\partial S(x,y)}{\partial c(x,y)} = 0. \tag{4}$$

From Eq. (4), omitting the coordinate (x,y), we get

$$\boldsymbol{\alpha} = \mathbf{M}^{-1}\boldsymbol{\beta}, \tag{5a}$$

where

$$\boldsymbol{\alpha} = [a,b,c]^{\mathrm{T}}, \tag{5b}$$

$$\boldsymbol{\beta} = [\sum_{m=1}^{M} I_m, \sum_{m=1}^{M} I_m\cos\Delta_m, \sum_{m=1}^{M} I_m\sin\Delta_m]^{\mathrm{T}}, \tag{5c}$$

and

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$$\mathbf{M}=\begin{bmatrix} M & \sum_{m=1}^{M}\cos\Delta_m & \sum_{m=1}^{M}\sin\Delta_m \\ \sum_{m=1}^{M}\cos\Delta_m & \sum_{m=1}^{M}\cos^2\Delta_m & \sum_{m=1}^{M}\cos\Delta_m\sin\Delta_m \\ \sum_{m=1}^{M}\sin\Delta_m & \sum_{m=1}^{M}\sin\Delta_m\cos\Delta_m & \sum_{m=1}^{M}\sin^2\Delta_m \end{bmatrix}. \tag{5d}$$

And wavefront phase at pixel (x,y) could be determined by

$$\varphi(x,y)=\tan^{-1}[-c(x,y)/b(x,y)]. \tag{6}$$

To make matrix **M** nonsingular, the interferogram number M should not be less than three.

Step 2. Determine tilt phase shifts in x-direction with wavefront phase and y-directional tilt phase shifts known

As shown in Fig. 1(a), each interferogram is X columns by Y rows. For pixels in the xth column of the mth interferogram, the x-directional phase shift, $\Delta_{xm}(x)=k_{xm}x+\delta_{xm}$, is uniform. And the phase shifts of the xth column are independent on y-directional tilt. After knowing phase shifts of X columns $\Delta_{xm}(x)$, (x = 1,2, …,X), in the mth interferogram, k_{xm} and δ_{xm} could be determined from a linear regression.

The method to calculate column phase shifts is described here. Like Wang's method, it's assumed that A and B are unvariable in each individual column, i.e. A and B are functions of x. And phase $\varphi(x,y)$ is known in Step 1 and k_{ym} and δ_{ym} are obtained in last iteration. For the pixels in the xth column of the mth interferogram, the theoretical intensity can be expressed as

$$I_m^{\,t}(x,y)=a'_m(x)+b'_m(x)\cos[\varphi'_m(x,y)]+c'_m(x)\sin[\varphi'_m(x,y)], \tag{7}$$

where $a'_m(x)=A(x)$, $b'_m(x)=B(x)\cos[\Delta_{xm}(x)]$, $c'_m(x)=-B(x)\sin[\Delta_{xm}(x)]$ remain unknown, and $\varphi'_m(x,y)=\varphi_m(x,y)+\Delta_{ym}(y)$ is known. The least-squares error of the xth column in the mth interferogram is

$$S'_m(x)=\sum_{y=1}^{Y}[I_m^{\,t}(x,y)-I_m(x,y)]^2. \tag{8}$$

According to least-squares criteria, applying partial derivative of $S'_m(x)$ with respect to $a'_m(x)$, $b'_m(x)$ and $c'_m(x)$ and making partial derivatives to be zero respectively, we get

$$\boldsymbol{\alpha}'_m(x)=\mathbf{M}'^{-1}_m(x)\boldsymbol{\beta}'_m(x), \tag{9a}$$

where

$$\boldsymbol{\alpha}'_m(x)=[a'_m(x),b'_m(x),c'_m(x)]^{\mathrm{T}}, \tag{9b}$$

$$\boldsymbol{\beta}'_m(x)=[\sum_{y=1}^{Y}I_m,\sum_{y=1}^{Y}I_m\cos\varphi'_m,\sum_{y=1}^{Y}I_m\sin\varphi'_m]^{\mathrm{T}}, \tag{9c}$$

and

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$$\mathbf{M}'_m(x)=\begin{bmatrix} Y & \sum_{y=1}^{Y}\cos\varphi'_m & \sum_{y=1}^{Y}\sin\varphi'_m \\ \sum_{y=1}^{Y}\cos\varphi'_m & \sum_{y=1}^{Y}\cos^2\varphi'_m & \sum_{y=1}^{Y}\cos\varphi'_m\sin\varphi'_m \\ \sum_{y=1}^{Y}\sin\varphi'_m & \sum_{y=1}^{Y}\sin\varphi'_m\cos\varphi'_m & \sum_{y=1}^{Y}\sin^2\varphi'_m \end{bmatrix}. \tag{9d}$$

In Eq. (9) coordinates (x,y) of I_m and φ'_m are omitted. The phase shift of the xth column in the mth interferogram could be determined from $\boldsymbol{\alpha}'_m(x)$ elements computation that is

$$\Delta_{xm}(x)=\tan^{-1}[-c'_m(x)/b'_m(x)]. \tag{10}$$

Then, phase unwrapping is needed to make $\Delta_{xm}(x)$ a continuous data set. Tilt gradient k_{xm} and translational phase shift δ_{xm} could be obtained by performing linear regression to $\Delta_{xm}(x)$, $(x = 1,2,\ldots,X)$.

Step 3. Determine tilt phase shifts in y-direction with wavefront phase and x-directional tilt phase shifts known

Determining gradient k_{ym} and translational phase shift δ_{ym} is similar to that in Step 2. The phase shift of the yth row in the mth interferogram is $\Delta_{ym}(y) = k_{ym}y + \delta_{ym}$ and is independent on x-directional tilt. Here, we assume that A and B are constants in each row. With wavefront phase $\varphi(x,y)$ and k_{xm} and δ_{xm} are known in last two steps, phase shifts of rows could be calculated with least-squares fitting method. For pixels in the yth row of the mth interferogram, $\varphi''_m(x,y)$ is known with denoting $\varphi''_m(x,y) = \varphi_m(x,y) + \Delta_{xm}(x)$. And three denotations, $a''_m(y) = A(y)$, $b''_m(y) = B(y)\cos[\Delta_{ym}(y)]$ and $c''_m(y) = -B(y)\sin[\Delta_{ym}(y)]$, are unknowns. Omitting the procedure of least-squares method, we get the least-squares equation

$$\boldsymbol{\alpha}''_m(y)=\mathbf{M}''^{-1}_m(y)\boldsymbol{\beta}''_m(y), \tag{11a}$$

where

$$\boldsymbol{\alpha}''_m(y)=[a''_m(y),b''_m(y),c''_m(y)]^{\mathrm{T}}, \tag{11b}$$

$$\boldsymbol{\beta}''_m(y)=[\sum_{x=1}^{X}I_m,\sum_{x=1}^{X}I_m\cos\varphi''_m,\sum_{x=1}^{X}I_m\sin\varphi''_m]^{\mathrm{T}}, \tag{11c}$$

and

$$\mathbf{M}''_m(y)=\begin{bmatrix} X & \sum_{x=1}^{X}\cos\varphi''_m & \sum_{x=1}^{X}\sin\varphi''_m \\ \sum_{x=1}^{X}\cos\varphi''_m & \sum_{x=1}^{X}\cos^2\varphi''_m & \sum_{x=1}^{X}\cos\varphi''_m\sin\varphi''_m \\ \sum_{x=1}^{X}\sin\varphi''_m & \sum_{x=1}^{X}\sin\varphi''_m\cos\varphi''_m & \sum_{x=1}^{X}\sin^2\varphi''_m \end{bmatrix}. \tag{11d}$$

In Eq. (11) coordinates (x,y) of I_m and φ''_m are omitted. The phase shift of the yth row in the mth interferograms is

$$\Delta_{ym}(y)=\tan^{-1}[-c''_m(y)/b''_m(y)]. \tag{12}$$

After performing phase unwrapping to $\Delta_{ym}(y)$, $(y = 1,2,\ldots,Y)$, tilt gradient k_{ym} and translational phase shift δ_{ym} could be determined by linear regression.

The three steps of iteration are shown in Fig. 1(b) for clarity. In Step 2 and 3, to make matrix $\mathbf{M}'$ and $\mathbf{M}''$ nonsingular, at least three pixels are required in a column and a row. And

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at least two columns and rows are required in each interferogram to determine gradients and translational phase shift by linear regressions. Therefore, at least three interferograms that contains three by three pixels are required with proposed method. And this requirement could be met easily. For practical use, interferograms of more pixels should be captured to make linear regressions be far from ill-conditions. And to make curvature matrix $\mathbf{M}'$ in Eq. (9d) and $\mathbf{M}''$ Eq. (11d) nonsingular, interferograms with at least one oblique fringe should be considered. Thus, the wavefront phases of pixels in each columns and rows are distributed over interval [0,2π], not confined to a small interval.

Proposed method is suitable for both rectangular and non-rectangular test surfaces. When dealing with non-rectangular pupils, the columns and rows of less than three pixels should be eliminated in iteration cycles. The phase-shift planes of entire pupil could be determined by residual columns and rows. And the wavefront phase of eliminated pixels could be also evaluated with known phase shifts.

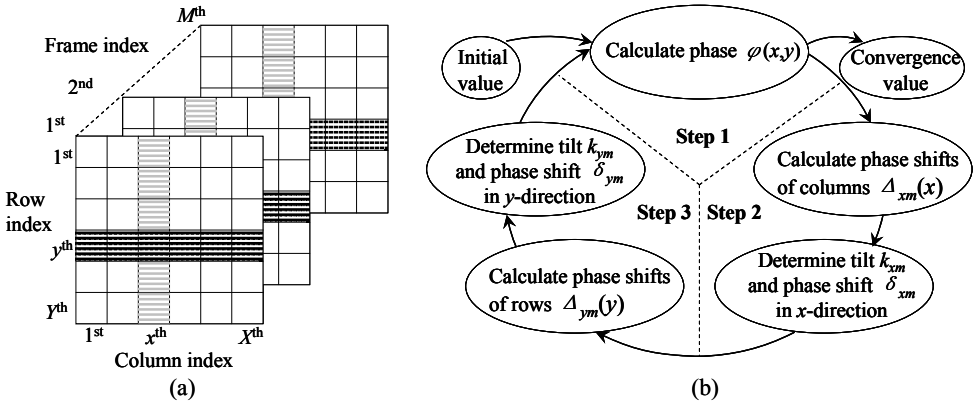


Fig. 1. (a) Schematic of tilt phase-shift interferograms. (b) Three steps in each iteration cycle.

The convergence criterion is composed of two parts that check the convergence errors of translational phase shifts and tilt gradients respectively. The sum of δ_{xm} and δ_{ym} is considered as translational phase shift of the mth interferogram. The convergence criterion is

$$\begin{cases} |[(\delta_{xm}^{i}+\delta_{ym}^{i})-(\delta_{x1}^{i}+\delta_{y1}^{i})]-[(\delta_{xm}^{i-1}+\delta_{ym}^{i-1})-(\delta_{x1}^{i-1}+\delta_{y1}^{i-1})]|<\varepsilon_1 \\ |(k_{xm}^{i}-k_{x1}^{i})-(k_{xm}^{i-1}-k_{x1}^{i-1})|+|(k_{ym}^{i}-k_{y1}^{i})-(k_{ym}^{i-1}-k_{y1}^{i-1})|<\varepsilon_2 \end{cases}. \tag{13}$$

The residual error thresholds ε_1 and ε_2 are preset values to check the relative changes between two adjacent iteration cycles. In our practical calculations, ε_1 and ε_2 are usually set to 10^{-5} and 10^{-6} respectively. The algorithm will converge after 5 to 15 iteration cycles. The number of iterations cycles usually depends on initial input value and convergence threshold values.

To summarize proposed method, the basic ideas are listed below. Firstly, tilt phase shifts are independent in x- and y-directions and could be separated. Secondly, for an interferogram, x-directional phase shift of each column is uniform and so is y-directional phase shift of each row. Thirdly, gradients and translational phase shifts could be determined by linear regressions.

3. Simulations

To evaluate the performance of proposed method, computer simulations were carried out. In our simulations, a spherical phase with slight tilt was chosen as test wavefront. We defined the wavefront phase as $\varphi(x,y) = 0.15\pi(x^2 + y^2) + \pi x/128 + \pi y/128$, the background as $A(x,y) = 140\exp[-0.2(x^2 + y^2)]$, and the modulation amplitude as $B(x,y) = 110\exp[-0.1(x^2 + y^2)]$ where $-1\le x\le 1$, $-1\le y\le 1$. The resolution of simulated interferograms was 128 by 128. And Gaussian white noise of variance 9 was added to interferogram intensity. Eight interferograms with tilt

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phase shifts were generated. With proposed method the tilt phase shifts were extracted and a comparison between real and extracted values is listed in Table 1. The initial values of tilts are zero and that of phase shifts are $\pi/2$ between adjacent interferograms. And the wavefront phase was extracted with Wang's method [11], Xu's method [13], and proposed method respectively, and the results are shown in Fig. 2. When performing Xu's method, the interferograms are divided to 4 by 4 regions. From Table 1 and Fig. 2, the accuracy of proposed method is proved. For there is no additional approximation error, proposed method could retrieval phase accurately. The PV value of phase retrieval errors is 0.0088λ with proposed method, while that are 0.149λ with Wang's method and 0.0430λ with Xu's method. Proposed method could alleviate fluctuation errors that are caused by tilt phase shifts. The maximum tilt in Table 1 is 4.266×10^{-2}rad/pixel, about 1.7-wave tilt over the entire pupil.

Additionally, a statistical simulation was conducted to study the precision of proposed method. Under above simulating conditions, 1000 frames of interferograms with Gaussian white noise added to were generated. And tilt gradients and translational phase shifts were extracted and the retrieval errors were calculated statistically. The results indicate, the standard deviation of phase shift retrieval error is 0.0218rad and that of tilt gradient retrieval error is 1.498×10^{-4}rad/pixel.

Table 1. Extracted Results of Tilt Phase Shifts[a]

Frame	k_x ($\times 10^{-2}$rad/pixel)		k_y ($\times 10^{-2}$rad/pixel)		δ (rad)	
	Real	Calculated	Real	Calculated	Real	Calculated
2	–0.515	–0.498	1.743	1.739	1	0.986
3	1.296	1.296	–1.076	–1.054	2	1.982
4	0.972	0.972	0.254	0.248	3	3.000
5	–0.187	–0.206	0.877	0.877	4	4.011
6	0.716	0.708	–1.753	–1.791	5	5.030
7	1.620	1.621	1.847	1.843	6	6.007
8	4.256	4.266	0.188	0.193	7	6.989

[a] Assuming that tilt phase shift of first interferogram is zero.

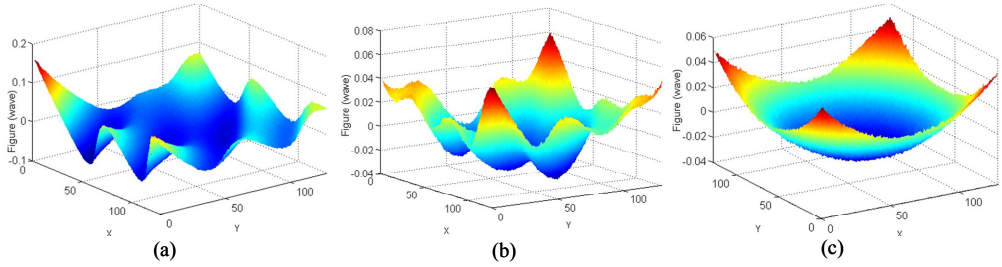


Fig. 2. Simulation results. The test wavefront is a spherical surface and its PV value is 0.075λ. Retrieved phases with Wang's method (a), Xu's method (b) and proposed method (c).

4. Experiments and discussions

Proposed method was implemented to test a flat surface and the performance was verified experimentally. A Fizeau interferometer fixed on a vibration-isolating platform was used to measure the surface of a flat. To reduce the blur of interferograms, the exposure time of camera was set to 2ms. And the resolution of recorded interferograms was 125 by 125 and gray depth is 256-levels. The preset value of phase shift was $-\pi/2$ between two adjacent interferograms. During the measurement, vibration was introduced by tapping the holder of flat with fingers. And six interferograms with tilt phase shifts was recorded and are shown in Fig. 3. There are four to five fringes in each interferogram. It is obvious that the tilts of last two interferograms are changed greatly, and it will be proved by the results of tilt gradients extracting in later discussions. For comparison, the phase of tested flat surface was measured without vibration and the result could be regarded as an exact value that is shown in Fig. 4. The PV value of the flat phase is 0.150λ. Three methods, Wang's, Xu's and proposed

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methods, were applied to interferograms in Fig. 3 to retrieve wavefront phase and results were shown in Fig. 5(a), Fig. 6(a) and Fig. 7(a) respectively. And the phase deviations from exact phase in Fig. 4, named phase retrieval errors, are exhibited in Fig. 5(b), Fig. 6(b) and Fig. 7(b) respectively. With Wang's method, for no tilt shifts are compensated, large tilt phase-shift errors are residual. For Wang's method, the PV value of phase retrieval error is 0.105λ and RMS value is 0.0120λ. With Xu's method, interferograms are divided to 4 by 4 regions and tilt phase shifts are compensated. The phase retrieval errors are alleviated but some obvious phase retrieval errors are residual. With Xu's method, the PV value of phase retrieval errors is 0.0394λ and RMS 0.00596λ. With proposed method, the tilt phase shifts are extracted and fluctuations caused by vibration are removed. The PV value of retrieval errors is 0.0277λ and the RMS is 0.00395λ by proposed method. The proposed method could suppress fluctuations introduced by vibration, and the residual errors may result from the detecting errors of tilt phase shifts.

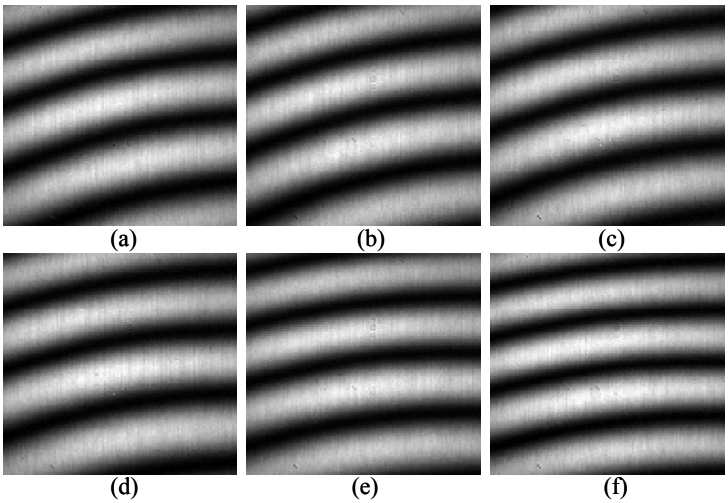


Fig. 3. (a)-(f) Recorded interferograms in the presence of vibrations. The preset phase shift value is -π/2 between adjacent interferograms.

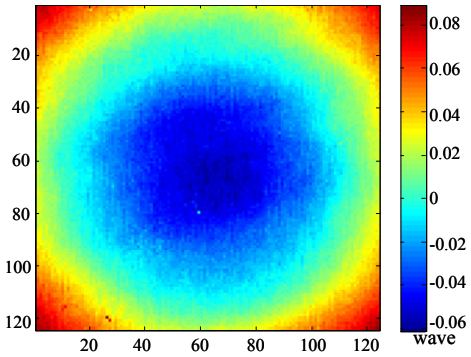


Fig. 4. Phase retrieved from interferograms without vibration. This phase value is regarded as exact value for comparison.

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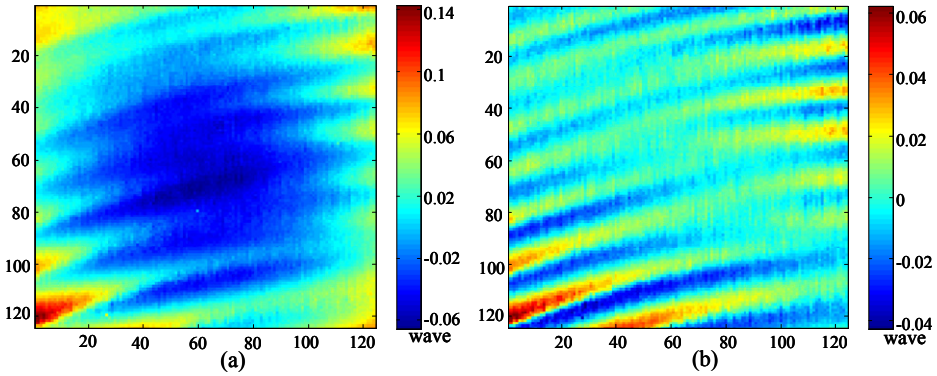


Fig. 5. (a) Phase retrieved from interferograms in Fig. 3 with Wang's method, and (b) phase retrieval errors compared with accuracy phase values in Fig. 4.

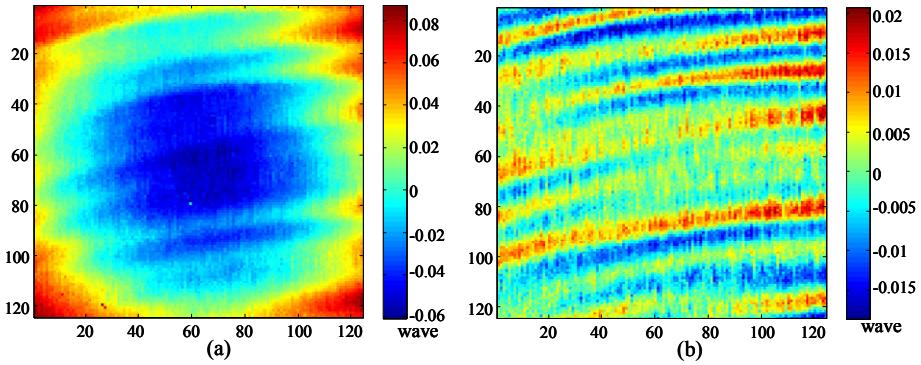


Fig. 6. (a) Phase retrieved from interferograms in Fig. 3 with Xu's method, and (b) phase retrieval errors compared with accuracy phase values in Fig. 4.

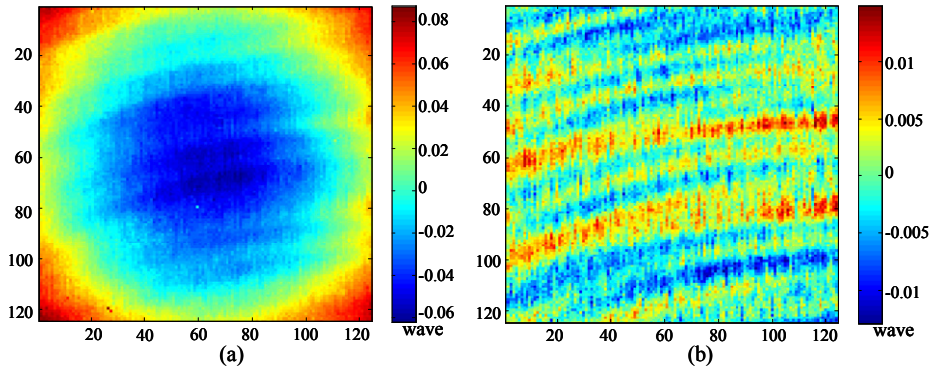


Fig. 7. (a) Phase retrieved from interferograms in Fig. 3 with proposed method, and (b) phase retrieval errors compared with accuracy phase values in Fig. 4.

Furthermore, tilt gradients and translational phase shifts were extracted to evaluate the performance of proposed method. From the extracting results, the largest deviation of phase shift from preset value $-\pi/2$ is almost π. The tilts of phase shift in x- and y-directions are shown in Fig. 8. The tilt phase shifts of first interferogram are assumed as zero and are subtracted from tilt phase shifts of other interferograms. The largest x-directional tilt gradient is 0.0157rad/pixel in the fifth interferogram and the largest y-directional tilt gradient is –0.0345rad/pixel in the sixth interferogram. The largest tilt of phase-shift planes is present in sixth interferogram and is 0.0375rad/pixel, about 1.5-wave wavefront tilt over the entire pupil.

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The tilt phase shift curves of columns or rows are believed to be a straight line, as shown in Fig. 8(a). However, waveness are present in *y*-directional tilt curves in Fig. 8(b). This phenomenon could be explained as a *printing-through* effect of fringes, for when calculating phase shifts with least-squares iterative algorithm [11], the backgrounds and modulation amplitudes are assumed unvariable in a column or a row. Therefore, the intensity of interferograms has some influence on extracted phase shift values. The interferograms we acquired in experiments contain fringes along *x*-direction, as shown in Fig. 3. Therefore, the sums of intensity in columns are almost same, while the sums of intensity in rows are variable due to the distribution of fringes. Hence, the extracted phase shifts of rows are wavy and the shape of waves are similar to fringes intensity along *y*-direction. However, the waveness hardly affect phase retrieval with propose method, for what we need are gradients and translational values of the curves.

In practice, convergence speed is another consideration besides accuracy. For comparison, the residual iterative error curves are shown in Fig. 9. For three methods, the initial input values are same and residual error threshold ε_1 and ε_2 are set to 10^{-6} and 10^{-7} respectively. As shown in Fig. 9, three methods have about the same convergence speed at first several iteration cycles. However, due to approximation error, Xu's method could not reach the given threshold. But proposed method converges to the given threshold after thirteen iteration cycles.

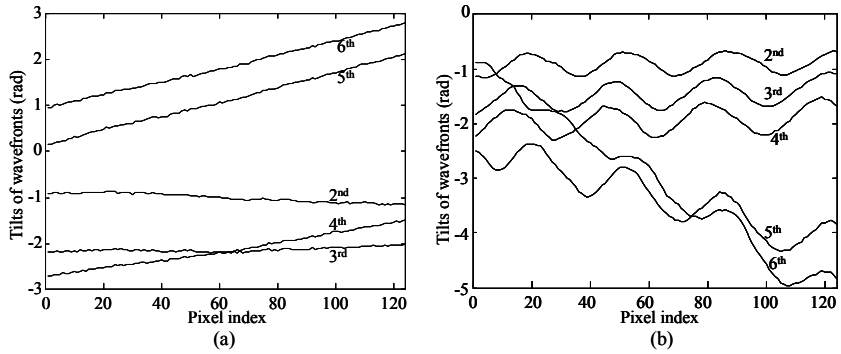


Fig. 8. Calculated wavefront tilts of interferograms in Fig. 3 in *x*- (a) and *y*-directions (b).

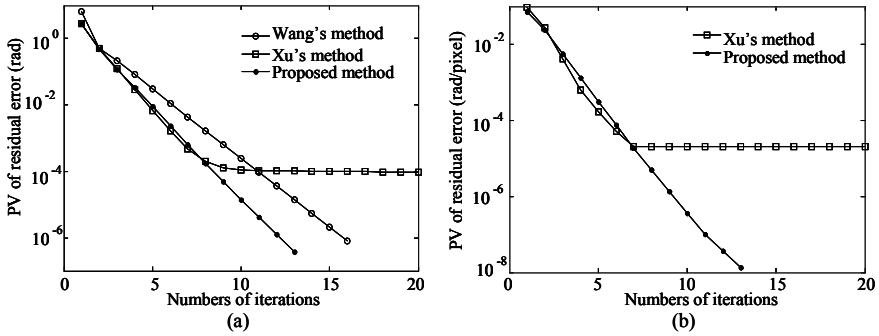


Fig. 9. Comparison of convergence speeds. Residual iteration errors in translational phase shifts calculations (a) and in tilt gradients calculations (b).

Computational efficiency is another consideration in practical use. For the phase shift of every row and column should be calculated, the cost time of proposed method is more dependent on size of interferograms than frame of interferograms. For six interferograms of 128 by 128 pixels, less than 30 seconds is needed to complete thirteen iterations with a 2.6GHz CPU. While computation for that of 512 by 512 pixels costs four minutes. When determining tilt gradient and translational phase shift, at least two columns or rows are needed

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to make linear regressions, therefore sampling with interval pixels are accepted to evaluate phase-shift planes. Computational efficiency could be heightened with little accuracy loss.

5. Conclusions

We have proposed a least-squares iterative method to deal with tilt phase shifts in temporal phase-shift interferometry. The basic ideas are that tilt phase shifts are composed of x- and y-directional tilt components and these two components are independent. Therefore, the x- and y-directional tilts could be extracted independently. Proposed method extracts phase shifts of columns and rows and makes linear regressions to determine gradients and translational phase shifts of interferograms. No additional approximation errors or nonlinear least-squares fitting are needed in our method. Therefore, it could retrieve phase accurately from tilt phase-shift interferograms. And it does not slowdown the iteration calculations. Hardware requirement is easy to meet for at least three frames of interferograms of three by three pixels are needed. Simulations and experiments demonstrate the performance of proposed method. The experimental results indicate that proposed method could be applied to interferometers in vibrating environment.

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